

Devansh Parapalli

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Professional Summary

Software engineer owning AI systems end-to-end, from requirements and architecture through to production. Engineered multiple production systems for Cisco and TCS, spanning support, finance, audit, with working range across full-stack services, data pipelines, and cloud architecture. The work performed directly maps to **1,000+ person-hours saved per month** and **100+ support cases assisted to correct closure each month**. Published in IJTE (2025). Certified GCP Professional Cloud Architect.

Work Experience

- Software Development Engineer**
Tata Consultancy Services Limited
- Jul 2025 — Present
Hyderabad, India
- Client: Cisco** – Defined requirements, owned architecture, and drove cross-functional alignment across Cisco support, FinOps, and audit teams as the lead (and sometimes solo) engineer and architect across multiple concurrent production engagements.
 - Support automation:** Cut support triage from a full business day to less than 30 seconds for multiple finance tracks within Cisco, by building APIs and Agentic AI workflows with multi-intent detection, tool-calling, MCP Servers, Kafka event streaming and auto-scaling runners; assists 100+ cases per month to correct closure. Currently extending it to Agentic AI-based autonomous resolution for all tracks.
 - Finance Automation:** Saved 1000+ person-hours a month across payroll review and FinOps reporting by replacing manual work with Agentic AI and ML-enabled pipelines over Snowflake, Oracle, MongoDB, plus a self-built DAG framework for extensible quarterly CFO fund-flow reporting on highly restricted data.
 - Data Reconciliation Platform:** Designed and built a reconciliation engine that turns plain-English steps into concrete processing and matching operations, capable of handling 100+ workflows at 1M+ rows per side.
 - Audit validation.** Built an independent audit workflow for an existing IDP pipeline, validating its output through a separate extraction methodology and flagging where the two disagree; calibrated with eval sets and human review, sampling 30% of global invoices daily (about 300 documents).
 - Architected orchestration per workload using custom LLM harnesses, MCP and FastMCP servers and callable AI Agents to expose internal data sources and tools as composable, reusable interfaces across pipelines.
 - GenAI enablement within the account: mentor 10+ engineers monthly on applying AI to their workflows.
 - Internal: GenAI Center of Excellence** – Led technical architecture and client demos for 10 RFPs, contributing to engagements spanning GenAI enablement, cloud migration, and legacy modernization
 - Architected an on-premises vLLM + LiteLLM platform with gateway routing, PII detection, RBAC, and audit logging for internal model serving with no external API dependency.
 - Delivered five company-wide Google Cloud training sessions and regularly support internal GenAI enablement initiatives.

Projects

- AIKO: AI-powered Knowledge Organizer** (aiko.parapalli.dev)
- Jun 2024 — Present
- Achieved **Mean Reciprocal Rank 0.75 / nDCG@5 0.7** over a 100K-chunk technical knowledge corpus by implementing hybrid retrieval (semantic search + keyword fallback) with tuned chunking and embedding selection across multi-modal content.
 - Sustained **p95 ingest latency of 3s** across a multi-modal document pipeline (PDFs, web content, multimedia) via a custom serialization layer with cloud-native and offline-capable deployment targets.
 - Reduced LLM response latency by **25%** by replacing SSE with custom **WebSocket** duplex streaming for real-time token delivery.
 - Built a custom orchestration framework for full control over prompt construction, retrieval, and tool-calling – replacing third-party abstractions with task-specific logic.

Education

- Government College of Engineering, Nagpur**
B.Tech, Computer Science and Engineering
CGPA: 8.28/10.0
- Nagpur, India
Dec 2021 — Jun 2025

Technical Skills

- AI & Inference:** LLM Serving (vLLM, LiteLLM), MLOps, LLM Evaluation, Multimodal Pipelines, XGBoost
- Agentic Systems:** Agentic AI, Multi-Agent (A2A), MCP / FastMCP, Tool Calling, Human-in-the-Loop, Google ADK
- Retrieval:** RAG, Hybrid Retrieval, Embeddings, Vector DBs (pgvector, FAISS, Pinecone)
- Backend & Systems:** System Design, Distributed Systems, Event-Driven Architecture (Kafka), Async Concurrency, FastAPI
- Data:** Data Pipelines, Snowflake, PostgreSQL, Oracle, MongoDB, SQL
- Cloud & Security:** GCP (Certified Architect), AWS, Docker, Kubernetes, Terraform, OpenShift, RBAC, PII Detection, Audit Logging
- Languages:** Python, TypeScript, C/C++, SQL

Certifications & Achievements

- GCP Certifications:** Professional Cloud Architect (Dec 2025) | Cloud Digital Leader (Sep 2025) | Generative AI Leader (Sep 2025)
- Publication:** “AI-based Knowledge Organizer for diverse data formats” - IJTE, March 2025